15A 1 CHIRALITY AND ENANTIOMERS

SPECIFICATION REFERENCE
15.1 15.2

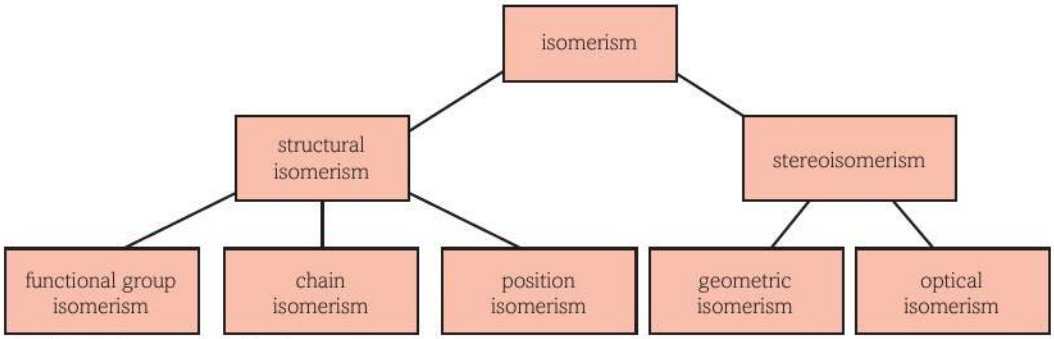
LEARNING OBJECTIVES

- Know that optical isomerism is a result of chirality in molecules with a single chiral centre.
- Understand that optical isomerism results from chiral centre(s) in a molecule with asymmetric carbon atom(s) and that optical isomers (enantiomers) are object and non-superimposable mirror images and be able to draw 3D diagrams of these optical isomers.

DIFFERENT TYPES OF ISOMERISM

In **Book 1**, you learned about structural isomerism and its division into chain isomerism and position isomerism. You then considered a different type of isomerism called 'geometric isomerism', including the terms *E*–*Z* and *cis*–*trans* isomerism – this is one type of stereoisomerism.

This chapter looks at a second type of stereoisomerism – optical isomerism. Before we introduce this, it is useful to consider all the types of isomerism that you need to be familiar with, in the form of a family tree.



▲ **fig A** The isomerism family tree.

Structural isomers are compounds with the same molecular formula but with different structural formulae.

Functional group isomers differ because they have different functional groups (e.g. aldehydes and ketones, alcohols and ethers).

Chain isomers differ because they have different patterns of branching in their carbon chains.

Positional isomers differ because the same functional group is attached to different carbon atoms in the chain.

Stereoisomers have the same structural formula but differ because their atoms or groups are arranged differently in three dimensions.

Geometric isomers differ because their atoms or groups are attached at different positions on opposite sides of a C=C double bond.

OPTICAL ISOMERISM

How does optical isomerism fit into this family tree? Optical isomers are non-superimposable mirror images of each other but are otherwise the same. This needs careful explanation!

CHIRALITY

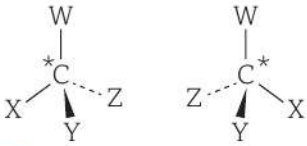
This is a term derived from the Greek word for 'hand', and could be translated as 'handedness'. Many objects, including human hands, can be described as **chiral**. Your hands have the same features, and you could describe both hands using the same words. Both consist of a palm with a thumb on one

side and four fingers of different sizes. However, if you place one hand on top of the other you can see that they are different – you cannot superimpose one hand on the other.

Now put your hands together – you can see that the thumbs are together, the little fingers are together, and so on – everything corresponds. Next, put your left hand in front of a mirror, then look at the image in the mirror and your right hand at the same time – they should look identical, so your hands can be described as mirror images, or sometimes as an object and its mirror image.

CHIRALITY IN SIMPLE MOLECULES

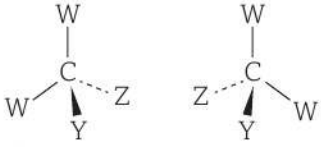
The key to understanding chirality in molecules is to visualise them in 3D, and ideally use models to help you. Think of a molecule consisting of a single carbon atom joined to four different groups or atoms (represented by W, X, Y and Z). If you represent the molecule in two dimensions, with bond angles of 90°, you cannot easily see why there should be two different arrangements. However, you may remember from **Topic 3 (Book 1: IAS)** on the shapes of molecules that you can attempt to show the 3D nature of molecules using different styles to represent the bonds. Ordinary lines are used to represent bonds in the plane of the paper, tapered or wedge-shaped lines indicate those above the plane of the paper, and dashed lines indicate those below the plane of the paper. Drawing the two structures side-by-side clearly shows how they relate to each other as object and mirror image (**fig B**).



▲ **fig B** This is one way to show optical isomers in 3D.

The asterisk (*) indicates that the carbon atom next to it is a chiral centre, also known as an **asymmetric** carbon atom.

If the attached atoms or groups are W, W, X and Y (so that two are the same) it is possible to show that they relate to each other in the same way (they are mirror images). However, there is no chirality because one can also be superimposed on the other (**fig C**).



▲ **fig C** These structures are not optical isomers because there are only three different groups joined to the central carbon atom.

HOW TO IDENTIFY CHIRAL CENTRES IN STRUCTURAL FORMULAE

You need to be able to identify chiral centres in molecules such as the examples shown in **table A**. You only need to consider whether there is a carbon atom joined to four different atoms or groups.

STRUCTURE	COMMENTS
CBr_2ClF	The single carbon atom has only three different atoms attached, so there is no chiral centre.
$CH_3CHBrCH_2CH_3$	Carbons 1, 3 and 4 in the chain have two or more hydrogens attached, so are not chiral. Carbon 2 is joined to four different groups (methyl, hydrogen, bromine and ethyl), so it is a chiral centre.
$CH_3CH_2CHBrCH_2CH_3$	Carbons 1, 2, 4 and 5 in the chain have two or more hydrogens attached, so are not chiral. Carbon 3 is joined to only three different groups (two ethyl, one hydrogen and one bromine), so it is not a chiral centre.
$CH_3CH_2CHBrCHBrCH_3$	Carbons 1, 2 and 5 in the chain have two or more hydrogens attached, so are not chiral. Carbon 3 is joined to four different groups (ethyl, hydrogen, bromine and $CHBrCH_3$), so it is a chiral centre. Carbon 4 is joined to four different groups (methyl, hydrogen, bromine and CH_3CH_2CHBr), so it is also a chiral centre.

table A

If a molecule has a chiral centre, then it can exist as optical isomers – each one is known as an **enantiomer**.

LEARNING TIP

It is really worthwhile using molecular models to make some models of simple structures to check the idea of mirror images and superimposability.

CHECKPOINT

1. What is the main similarity and the main difference between optical isomers and geometric isomers?
2. How many chiral centres are there in each of these molecules?
   (a) $CH_3CCl_2CH_3$
   (b) $CH_3CH(OH)COOH$
   (c) $CH_2Cl–CHCl–CHFCH_3$

SUBJECT VOCABULARY

chiral a chiral atom in a molecule is one that allows it to exist as non-superimposable forms. It can also refer to the molecule itself
asymmetric an asymmetric carbon atom in a molecule is one that is joined to four different atoms or groups
enantiomers isomers that are related as object and mirror image

15A 2 OPTICAL ACTIVITY

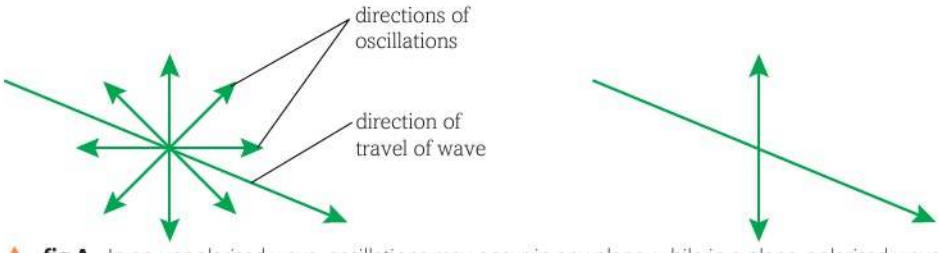
SPECIFICATION REFERENCE
15.3 15.4

LEARNING OBJECTIVES

- Know that optical activity is the ability of a single optical isomer to rotate the plane of polarisation of plane-polarised monochromatic light in molecules containing a single chiral centre.
- Know what is meant by the term 'racemic mixture'.

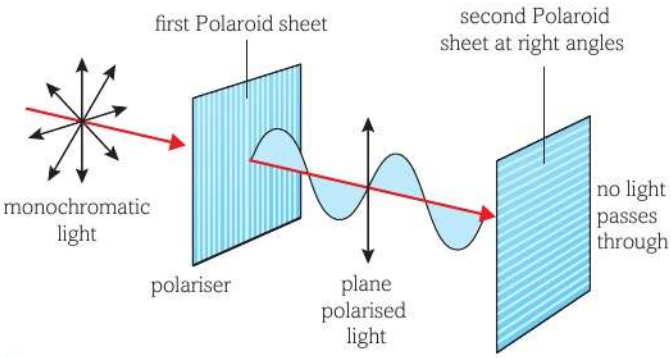
PLANE-POLARISED LIGHT

In the previous section we looked at the causes of optical isomerism. Before we consider **optical activity**, we must have some understanding of **plane-polarised light**. One way to consider light is as electromagnetic radiation that travels as a transverse wave. This means that the oscillations exist in planes at right angles to the direction of travel. **Fig A** shows 'normal' **unpolarised light** and plane-polarised light with the oscillations in only one plane.



▲ **fig A** In an unpolarised wave, oscillations may occur in any plane, while in a plane-polarised wave they occur in only one plane.

Some natural materials, and synthetic materials such as Polaroid (the material used in some sunglasses), can absorb all of the oscillations except those in a single plane. This means that they convert unpolarised light into plane-polarised light. For convenience, the single plane that remains is often assumed to be the vertical plane, as shown in **fig B**.



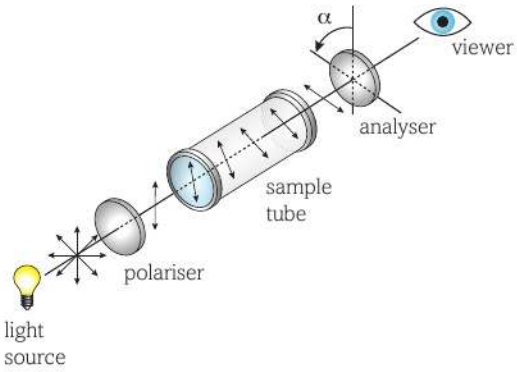
▲ **fig B** Polarisation of light.

Fig B shows that the second sheet of Polaroid has horizontal lines, which means that only horizontally plane-polarised light can pass through. As the first sheet has produced vertically plane-polarised light, then no light can pass through the second sheet.

POLARIMETRY

Polarimetry is the use of a **polarimeter** to measure the amount of optical activity, if any, of a substance. **Fig C** shows how a polarimeter works. A monochromatic light source (light of only one colour or frequency) passes through a polarising filter (such as a piece of Polaroid). This is called the **polariser** because it converts unpolarised light into (vertically) plane-polarised light. The plane-polarised light then passes through a sample tube containing some of the substance in solution. If the substance is optically active (because it contains an enantiomer), then the plane of polarisation

will be rotated so that it is no longer vertical. In this example, the rotation is anticlockwise, and the substance can therefore be described as 'laevorotatory'. If the rotation is clockwise, then the substance is 'dextrorotatory'.



▲ **fig C** You can see how the plane of polarisation of the polarised light is rotated anticlockwise as it passes through the sample tube.

The second polarising filter, known as the **analyser**, is rotated to a position where the maximum light intensity can be seen. The angle of rotation (α) is measured, and is quoted as a positive value if the rotation is clockwise and as a negative value if the rotation is anticlockwise. In **fig C**, the rotation, α, is approximately $-90°$.

PROPERTIES OF ENANTIOMERS

Two enantiomers have identical physical properties, with one exception. The enantiomers rotate the plane of polarisation of plane-polarised light by equal angles, but in opposite directions. So, if the rotation for one enantiomer is $+60°$, then the rotation for the other enantiomer is $-60°$.

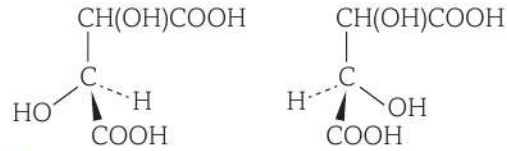
Two enantiomers have identical chemical properties, with one exception – the way in which they react with enantiomers of other substances. This chemical property may be different for each enantiomer.

RACEMIC MIXTURES

What happens if a compound with a chiral centre is present as a mixture of both enantiomers? You can imagine that the dextrorotatory and laevoratatory enantiomers have equal but opposite effects on plane-polarised light. This means that the analyser does not need to be rotated to allow the maximum light intensity to be seen. Without knowing anything about the substance, it is not possible to distinguish between a substance that has no optical activity and one that has optically active enantiomers whose effects cancel out. A mixture containing equal amounts of two enantiomers is called a **racemic mixture**.

We will briefly explain here the origin of this scientific term. The Latin word *racemus* means a bunch of grapes. How is this related to optical activity? The French chemist Louis Pasteur achieved a lot in his lifetime (1822–1895), and his name lives on in the word 'pasteurisation', often associated with milk production. He also realised that tartaric acid produced during the fermentation of grapes showed optical activity, but that tartaric acid produced in other ways did not. He eventually realised that a single enantiomer of tartaric acid was produced from fermenting grapes and that a mixture of both enantiomers of tartaric acid was produced in the other ways. Because tartaric acid is present in grapes, the term 'racemic acid' was used, and the 'racemic' part of this term was then used to describe an equimolar mixture of the two enantiomers.

Here are the structures of the two enantiomers of tartaric acid:



▲ **fig D** The two enantiomers of tartaric acid.

EXAM HINT

You may be asked to **draw** optical isomers in an exam. Make sure that you **identify** the chiral carbon with a * and use the wedge and dotted line notation.

LEARNING TIP

It is worth spending some time making sure you understand the differences between these terms:

unpolarised, polariser, plane-polarised, plane of polarisation.

CHECKPOINT

1. Outline how a polariser works.

SKILLS REASONING

2. A dextrorotatory enantiomer has a rotation of +43°. A mixture of this enantiomer and its laevorotatory enantiomer has a rotation of −10°. What does this information indicate about the composition of this mixture?

SUBJECT VOCABULARY

optical activity a substance shows optical activity if it rotates the plane of polarisation of plane-polarised light

plane-polarised light monochromatic light that has oscillations in only one plane

unpolarised light light that has oscillations in all planes at right angles to the direction of travel

polarimeter the apparatus used to measure the angle of rotation caused by a substance

polariser a material that converts unpolarised light into plane-polarised light

analyser a material that allows plane-polarised light to pass through it

racemic mixture an equimolar mixture of two enantiomers that has no optical activity

15A 3 OPTICAL ACTIVITY AND REACTION MECHANISMS

SPECIFICATION REFERENCE
15.5 PART

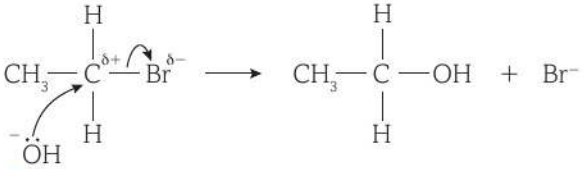
LEARNING OBJECTIVES

- Be able to use data on optical activity of reactants and products as evidence for S_N1 and S_N2 mechanisms.

NUCLEOPHILIC SUBSTITUTION IN HALOGENOALKANES

In **Book 1** you learned to write mechanisms for these reactions, but this was limited. We suggested that the explanation given in **Book 1** was incomplete and that there was more to come.

Now is the appropriate time to develop the explanation in greater depth. First, let us revise what you may remember from **Book 1**. **Fig A** shows how we represented the mechanism of the reaction between the primary halogenoalkane bromoethane and hydroxide ions.



▲ **fig A** This is a simple way to represent a nucleophilic substitution mechanism.

We must now accept that this mechanism has been made too simple and that we need to consider two alternative mechanisms. This mechanism shows the movement of two pairs of electrons happening at the same time – those from the O of OH^- ions and those from the C—Br bond. In fact, there are two distinct mechanisms.

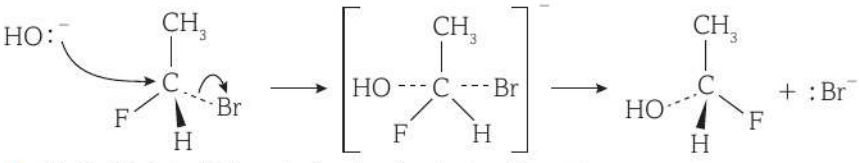
THE S_N2 MECHANISM

The individual symbols in 'S_N2' have the following meanings.

S = substitution N = nucleophilic 2 = **bimolecular**

The term 'bimolecular' means that there are two species involved in the rate-determining step – in this example, the OH^- ion and the halogenoalkane.

Bimolecular reactions were considered in **Topic 11** in this book. In this topic, we are going to revisit the mechanism provided in **Book 1**, by looking at it in more detail. Here is the mechanism for a primary halogenoalkane that is also an enantiomer.



▲ **fig B** This is the S_N2 mechanism showing the transition state.

Notice that the curly arrows are used in the same way. The mechanism now includes a transition state showing the HO—C bond forming and the C—Br bond breaking. The dashed lines represent partial bonds – those being broken and formed. The most important thing to notice is that the arrangement of the three groups that remain attached to the central carbon atom has been inverted (turned inside out).

LEARNING TIP

Remember to include square brackets and a negative charge outside the brackets when writing the transition state in the S_N2 mechanism. Using $\delta+$ and $\delta-$ to indicate the polar bonds is good practice, but in this section, they have been left out for clarity.

EXAM HINT

Now is a good time to make sure you know the exact definitions for nucleophile and electrophile.

EVIDENCE FROM OPTICAL ACTIVITY

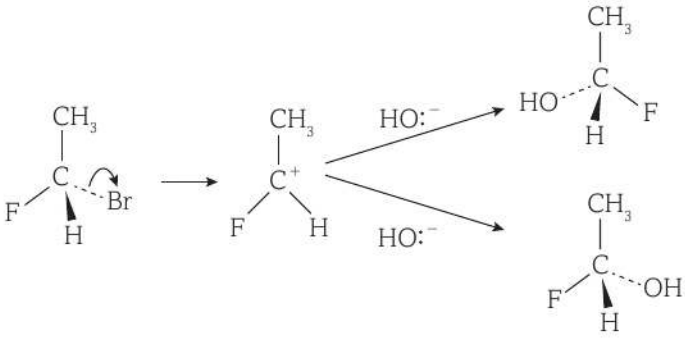
If the reaction actually occurs according to the S_N2 mechanism, then the optical activity of the product is different from that of the reactant. (This is because of the inversion that occurs.) If the reactant is dextrorotatory, then the product will be laevorotatory, and *vice versa*. So, by measuring the optical activity of the original halogenoalkane and the alcohol formed, we can show whether the reaction has occurred by the S_N2 mechanism.

THE S_N1 MECHANISM

The individual symbols in 'S_N1' have the following meanings.

S = substitution N = nucleophilic 1 = **unimolecular**

The term 'unimolecular' means that only one species is involved in the rate-determining step – in this example, just the halogenoalkane. In **fig C** we have used the same halogenoalkane as for the S_N2 mechanism so that you can focus on the differences between the two mechanisms.



▲ **fig C** In the S_N1 mechanism, the planar carbocation can be attacked from the left or from the right, giving a pair of enantiomers.

EXAM HINT

When showing the hydroxide ion as a nucleophile, make sure you indicate the lone pair and the negative charge on the oxygen atom. Do not show it as OH–.

The original halogenoalkane has a tetrahedral shape, but the product of the first step of the reaction is a carbocation with a planar shape. This means that in the second step of the reaction, there is an equal chance that the attacking HO^- ion can approach from the left or from the right. There are therefore two products, both of which are enantiomers. These are present in equal numbers, so a racemic mixture is formed.

EVIDENCE FROM OPTICAL ACTIVITY

If the reaction actually occurs according to the S_N1 mechanism, then the optical activity of the product is different from that of the reactant. Whether the reactant is dextrorotatory or laevorotatory, the product has no optical activity. So, by measuring the optical activity of the original halogenoalkane and the alcohol formed, we can show whether the reaction has occurred by this mechanism.

WHICH MECHANISM?

In general, the main mechanism for primary halogenoalkanes is S_N2, and for tertiary halogenoalkanes is S_N1. For secondary halogenoalkanes, both mechanisms can occur, although the main one depends on the actual halogenoalkane and the conditions.

CHECKPOINT

SKILLS DECISION-MAKING

1. 2-Bromobutane and cyanide ions react together by an S_N2 mechanism. Draw the structure of the transition state formed in this reaction.
2. 2-Bromopropane and cyanide ions react together by an S_N1 mechanism. Explain why there is no change in optical activity in this reaction.

SUBJECT VOCABULARY

bimolecular (mechanism) a mechanism in which two species are reacting in the rate-determining step

unimolecular (mechanism) a mechanism in which only one species is reacting in the rate-determining step